

BERYLLIUM

(Data in metric tons, beryllium content, unless otherwise specified)

Domestic Production and Use: One company in Utah mined bertrandite ore and converted it, along with imported beryl, into beryllium hydroxide. Some of the beryllium hydroxide was shipped to the company's plant in Ohio, where it was converted into metal, oxide, and downstream beryllium-copper master alloy, and some was sold. Estimated beryllium apparent consumption in 2024 was 170 tons and was valued at about \$260 million based on the most recent beryllium price estimate. Based on sales revenues, approximately 20% of beryllium products were used in industrial components, 19% in aerospace and defense applications, 11% in automotive electronics, 8% in telecommunications infrastructure, 6% each in consumer electronics and energy applications, 2% in semiconductor applications, and 28% in other applications. Beryllium alloy strip and bulk products, the most common forms of processed beryllium, were used in all application areas. Most unalloyed beryllium metal and beryllium composite products were used in defense and scientific applications. The U.S. Department of Defense supports the availability of domestic beryllium to meet critical defense needs. In 2010, under the Defense Production Act, Title III program, a public-private partnership with the leading U.S. beryllium producer reestablished domestic production of beryllium metal.

Salient Statistics—United States:

	2020	2021	2022	2023	2024^e
Production, mine shipments	165	175	175	185	180
Imports for consumption ¹	48	49	39	25	20
Exports ²	25	30	61	68	30
Shipments from Government stockpile ³	3	7	9	NA	NA
Consumption:					
Apparent ⁴	196	196	187	144	170
Reported, ore	170	170	170	180	180
Price, annual average unit value, beryllium-copper master alloy, ⁵	620	680	660	1,400	1,500
dollars per kilogram of contained beryllium					
Stocks, ore, industry, yearend	30	35	10	10	10
Net import reliance ⁶ as a percentage of apparent consumption	16	11	6	E	E

Recycling: Beryllium was recovered from new scrap generated during the manufacture of beryllium products and from old scrap. Detailed data on the quantities of beryllium recycled were not available but may account for as much as 20% to 25% of total beryllium consumption. The leading U.S. beryllium producer managed a recycling program for all its beryllium products, recovering approximately 40% of the beryllium content of the new and old beryllium alloy scrap.

Import Sources (2020–23):¹ Kazakhstan, 39%; Latvia, 23%; Japan, 17%; Canada, 6%; and other, 15%.

Tariff:	Item	Number	Normal Trade Relations 12–31–24
	Beryllium ores and concentrates	2617.90.0030	Free.
	Beryllium oxide and hydroxide	2825.90.1000	3.7% ad valorem.
	Beryllium-copper master alloy	7405.00.6030	Free.
	Beryllium-copper plates, sheets, and strip:		
	Thickness of 5 millimeters (mm) or more	7409.90.1030	3% ad valorem.
	Thickness of less than 5 mm:		
	Width of 500 mm or more	7409.90.5030	1.7% ad valorem.
	Width of less than 500 mm	7409.90.9030	3% ad valorem.
	Beryllium:		
	Unwrought, including powders	8112.12.0000	8.5% ad valorem.
	Waste and scrap	8112.13.0000	Free.
	Other	8112.19.0000	5.5% ad valorem.

Depletion Allowance: 22% (domestic), 14% (foreign).

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Government Stockpile:⁷

	FY 2024		FY 2025	
<u>Material</u>	<u>Potential acquisitions</u>	<u>Potential disposals</u>	<u>Potential acquisitions</u>	<u>Potential disposals</u>
Metal (all types)	—	7	—	7

Events, Trends, and Issues: Apparent consumption in 2024 increased by 18% from that in 2023 owing primarily to a 56% decrease in estimated beryllium exports offset by a 20% decrease in estimated imports. The decrease in exports reflected a large reduction in beryllium metal exports to Canada, Germany, Japan, and Mexico. The decrease in imports reflected a reduction in beryllium metal imports from Belgium, Kazakhstan, and Latvia. During the first 6 months of 2024, the leading U.S. beryllium producer reported that net sales of its beryllium alloy strip and bulk products and beryllium metal and composite products were 4% lower than those during the first 6 months of 2023. Net sales of beryllium products decreased primarily in the automotive, industrial components, and telecommunications and data markets. Because of the toxic nature of beryllium, various international, national, and State guidelines and regulations have been established regarding beryllium in air, water, and other media. Industry is required to carefully control the quantity of beryllium dust, fumes, and mists in the workplace.

World Mine Production and Reserves:

	Mine production ^{8, 9}		Reserves ¹⁰
	<u>2023</u>	<u>2024^e</u>	
United States	185	180	The United States has very little beryl that can be economically hand sorted from pegmatite deposits. An epithermal deposit in the Spor Mountain area in Utah is a large bertrandite resource, which is being mined. Proven and probable bertrandite reserves in Utah total about 19,000 tons of beryllium content. World beryllium reserves were not available.
Brazil	^e 40	80	
China	^e 74	77	
Madagascar	^e 1	1	
Mozambique	23	24	
Rwanda	^e 1	1	
Uganda	^e 1	1	
World total (rounded)	325	360	

World Resources:¹⁰ The world's identified resources of beryllium have been estimated to be more than 100,000 tons. About 60% of these resources are in the United States; by tonnage, the Spor Mountain area in Utah, the McCullough Butte area in Nevada, the Black Hills area in South Dakota, the Sierra Blanca area in Texas, the Seward Peninsula in Alaska, and the Gold Hill area in Utah account for most of the total.

Substitutes: Because the cost of beryllium is high compared with that of other materials, it is used in applications in which its properties are crucial. In some applications, certain metal matrix or organic composites, high-strength grades of aluminum, pyrolytic graphite, silicon carbide, steel, or titanium may be substituted for beryllium metal or beryllium composites. Copper alloys containing nickel and silicon, tin, titanium, or other alloying elements or phosphor bronze alloys (copper-tin-phosphorus) may be substituted for beryllium-copper alloys, but these substitutions can result in substantially reduced performance. Aluminum nitride or boron nitride may be substituted for beryllium oxide.

^eEstimated. E Net exporter. NA Not available. — Zero.

¹Includes estimated beryllium content of imported ores and concentrates, oxide and hydroxide, unwrought metal (including powders), beryllium articles, waste and scrap, beryllium-copper master alloy, and beryllium-copper plates, sheets, and strip.

²Includes estimated beryllium content of exported unwrought metal (including powders), beryllium articles, and waste and scrap.

³Change in total inventory from prior yearend inventory. If negative, increase in inventory. Beginning in 2023, Government stock changes no longer available.

⁴Defined for 2020–22 as production + imports – exports ± adjustments for Government and industry stock changes. Beginning in 2023, Government stock changes no longer included.

⁵Calculated from gross weight and customs value of imports; beryllium content estimated to be 4%. Rounded to two significant figures.

⁶Defined for 2020–22 as imports – exports ± adjustments for Government and industry stock changes. Beginning in 2023, Government stock changes no longer included.

⁷See Appendix B for definitions.

⁸In addition to the countries listed, Kazakhstan and Portugal may have produced beryl ore, but available information was inadequate to make reliable estimates of output. Other nations that produced gemstone beryl ore may also have produced some industrial beryl ore.

⁹Based on 4% beryllium content of bertrandite and beryl sources.

¹⁰See Appendix C for resource and reserve definitions and information concerning data sources.